PH101- Applied Physics

Pre-Requisite: None
Instructor: Mazhar Javed
Office: G-16, FBS
Email: mazhar@giki.edu.pk
(Office hrs: 10:00 am – 10:30 am, Monday, Tuesday, Wednesday Thursday)

Course Introduction

This course covers the fundamentals of engineering applying the concepts of basic mathematics and physics but at advanced level. The aim of this subject is to develop an understanding of real-world engineering applications and problem solving.

Course Contents

Proposed:

- Vector Operations, force vectors
- Motion in one and two dimensions
- Newton Laws and its applications
- Momentum and its conservation
- Rotational kinematics and dynamics
- Work and energy
- Coulomb's law and electric field calculation
- Gauss's law and electric field calculation

Mapping of Class Learning Outcome (CLOs) to Program Learning Outcomes (PLOs)

S. No	CLOs	PLOs	BT Level
	After completing this course, the student will be able to…		
CLO1	Apply vector operations, Newton's laws to problems of translational motion and rotational motion	PLO1	C3
CLO2	Apply fundamental conservation laws (momentum, angular momentum, work done and energy) to the problems of mechanics	PLO1	C3
CLO3	Apply the fundamental equations and concepts of electricity and magnetism to solve related problems	PLO1	C3

Grading Policy (Subject to change upon the discretion of the instructors)

Assessment Items….		**% Marks**
1.	Assignment	10%
2.	Quizzes	20%
3.	Mid-Term Exam	30% (After 8th week)
4.	Final Exams	40% (After 15th week)

Text and Reference Books

Textbooks:

1. *Hugh D. Young & Roger A. Freedman, "University Physics", 15th Edition, 2018*

2. *R.Resnick, D.Halliday & K.S. Krane, "Physics Volume-1", 5th Edition, 2002*
3. *R.Resnick, D.Halliday & K.S. Krane, "Physics Volume-2", 5th Edition, 2002*
4. *R.C. Hibbeler; Engineering mechanics, Statics, 14th Edition, Pearson Education*

Reference *books:*

1. *Frederick J.Keller,W.Edward Gettys & Malcolm J.Skove, "Physics: Classical and Modern"*
2. *Raymond A. Serway and John W.Jewett, "Physics for Scientists and Engineers", 6th Edition*
3. *Feynmann Lectures on Physics*

Administrative Instruction

- Student Attendance is expected to be 100%, and a minimum of 80% (mandatory) attendance is required to sit in the final exams.
- Students must pay attention to reading the textbooks chapter for course assessment.
- All the direct assessment tools i.e., Quizzes, Assignment, Midterms, Project and Final Exams must be attempted. Failure to attempt any of the assessment tools without any medical reasons may result in failure in that assessment. All quizzes would be taken together with other sections.
- Students are advised to study the previous lecture before the next class for better understanding.
- Class participation is highly encouraged. It develops more interest. Students are also advised to spare some time for group discussion with their classmates to explore new ideas.
- Handouts and related notes will be available on MS internet course portal.

For any query, please contact instructor during office time.